

Smart Vision Assistant: Hardware & Wiring Made Simple

An easy-to-read, plain-English explanation of all the physical parts, how they connect together, and how to safely build the wearable device without electrical confusion.

1. The Big Picture: What Are We Building?

Imagine having a friendly helper sitting on your chest or shoulder that acts as a **smart pair of eyes with a voice**. If you cannot see well, walking down the street or through a room can be scary because of hidden stairs, open doors, chairs, poles, or vehicles. Our device watches what is in front of you, measures how close obstacles are, and speaks out loud into your ear in real time (for example: *'Staircase directly ahead at 1 meter. Chair to your left.'*).

To make this work in real life, we only need **7 simple physical parts** that you can plug together on a workbench:

2. Meet the 7 Physical Parts (In Plain Words)

Part 1: The Brain — Raspberry Pi 5 — *Role: The Main Computer (Processes everything)*

What it does: It is a complete, powerful computer the size of a deck of cards. It takes live video frames from the camera, runs the artificial intelligence software to recognize objects, reads distance numbers, and generates spoken words.

Why we use it: It is cheap (\$60–\$80), fits in a pocket, and has modern math shortcuts (called ARM NEON) that let it analyze 15 to 20 video pictures every second without needing an expensive, heavy computer.

Part 2: The Eyes — Camera Module 3 or USB Webcam — *Role: Visual Sensor (Takes live pictures)*

What it does: Captures smooth video of your path at 30 pictures per second. It sends these pictures directly into the Raspberry Pi's memory so the AI can inspect them.

Why we use it: The Camera Module 3 has automatic focus (autofocus) so both close-up objects (like a coffee cup on a desk) and far-away objects (like a car or bus) look sharp and clear.

Part 3: The Bat Ears — HC-SR04 Ultrasonic Distance Sensor — *Role: Sound-Wave Radar (Measures physical distance)*

What it does: It has two silver cylinders that look like eyes. One is a tiny speaker that shoots out silent high-pitched sound waves; the other is a tiny microphone that listens for the bounce. By timing how long the echo takes to return, it knows the exact distance to obstacles in front of you.

Why we use it: Cameras can be fooled by clean glass doors (which look invisible) or dark hallways (where pictures turn black). Sound waves bounce off glass perfectly and work in total pitch darkness!

Part 4: The Electric Shield — Two Tiny Resistors (1kΩ and 2kΩ) — *Role: Voltage Safety Guard (Protects the computer from frying)*

What it does: The distance sensor talks using 5.0 Volts of electricity. But the Raspberry Pi can only safely listen to 3.3 Volts! If you connect 5 Volts directly into the Pi, it will fry the pin like a blown lightbulb. These two tiny 10-cent resistors act like a gentle water valve, safely stepping 5 Volts down to a comfortable 3.3 Volts.

Why we use it: It costs practically nothing, never breaks, adds zero delay, and completely protects your \$60 computer from getting damaged.

Part 5: The Question Button — Tactile Push Button — *Role: Manual Trigger (Ask for an instant overview)*

What it does: A small clicky button. Whenever you want to know what is around you right this second, just press it. The device immediately does a complete scan and tells you everything in your path.

Why we use it: Gives the user complete control. You don't have to wait for an obstacle to trigger an alert; you can ask whenever you feel unsure of your surroundings.

Part 6: The Voice — Mini Speaker or Headphones — *Role: Acoustic Feedback (Speaks aloud)*

What it does: Speaks natural warning messages directly into your ear using non-blocking software, keeping your hands totally free.

Why we use it: Plugs into any standard 3.5mm headphone jack or USB port. Bone-conduction headphones can also be used so your ears remain open to ambient street traffic sounds.

Part 7: The Fuel — Official 27W USB-C Power Supply — Role: *Power Delivery (Keeps everything running)*

What it does: Provides steady 5.1 Volts at 5 Amps. Can be plugged into a wall outlet or an approved high-power portable power bank.

Why we use it: Running 4 computer cores, a camera, and sensors at the same time takes real energy. A cheap, weak phone charger can cause the computer to suddenly shut down while you are walking.

3. The 40-Pin Connection Cheat Sheet

The Raspberry Pi 5 has 40 metal pins on its edge. You only need to plug wires into **6 of these pins**. Here is the complete, easy-to-follow table:

Physical Pin #	Header Label	Wire Color	Where It Plugs In	What It Does (Simple Words)
Pin 2	5V Power	Red Wire	Sensor VCC pin	Gives 5 Volts of electric power to turn the distance sensor ON.
Pin 6	Ground (GND)	Black Wire	Sensor GND pin	Completes the electrical circuit for the distance sensor (0 Volts).
Pin 16	GPIO 23	Yellow Wire	Sensor TRIG pin	The Pi sends a tiny pulse telling the sensor: 'Chirp sound now!'
Pin 18	GPIO 24	Green Wire	Middle of the 2 Resistors	Listens to the safe 3.3V echo sound pulse to calculate distance.
Pin 11	GPIO 17	Blue Wire	Button Terminal 1	Detects when you press the clicky button to ask for a scan.
Pin 9	Ground (GND)	Black Wire	Button Terminal 2	When button is pressed, it connects Pin 11 to Ground.
Pin 14	Ground (GND)	Black Wire	End of 2kΩ Resistor	Connects the voltage divider to Ground for safety.
Camera Port	CAM1 Connector	Ribbon Cable	Camera Module 3	Transfers high-speed live video pictures directly to computer memory.
USB / Audio	USB Port / Jack	Audio Cable	Speaker / Headphones	Plays the spoken voice warnings aloud.

REMINDER: THE RESISTOR VOLTAGE DIVIDER RULE

Never connect the sensor's **ECHO** pin straight to the Raspberry Pi! Always put the 1kΩ resistor in between, and connect the 2kΩ resistor down to Ground. Think of it like putting sunglasses on the Raspberry Pi so the bright 5V electric light doesn't hurt its eyes.

4. 5 Easy Steps to Wire Everything Up

- Step 1: Safety First — Unplug Power!
- Always make sure the Raspberry Pi is turned OFF and unplugged before touching any wires. This prevents accidental static sparks.
- Step 2: Connect Power & Ground Rails
- Plug a Red wire from Pin 2 (5V) to your breadboard's Red strip. Plug a Black wire from Pin 6 (GND) to your breadboard's Blue strip.
- Step 3: Plug in the Distance Sensor
- Put the HC-SR04 sensor into the breadboard. Wire its VCC pin to Red (+5V) and its GND pin to Blue (GND). Wire its TRIG pin to Pin 16 with a Yellow wire.
- Step 4: Add the Two Resistors (The Shield)
- Connect the 1kΩ resistor to the sensor's ECHO pin. Connect its other leg to the 2kΩ resistor. Connect the bottom leg of the 2kΩ resistor to Ground. Connect a Green wire from the middle spot between the two resistors straight to Pin 18.
- Step 5: Connect Button, Camera, and Speaker
- Plug the button across the breadboard. Connect one side to Pin 11 (Blue wire) and the other side to Ground (Black wire). Click the camera ribbon into the CAM1 slot, plug in your speaker, and you are ready to power on!

5. Simple Tests & Troubleshooting (If Something Goes Wrong)

What is Wrong?	What It Means (Root Cause)	How to Fix It (Simple Steps)
Distance reads 0.0m or gives errors	The echo signal is not reaching Pin 18.	Check your two resistors. Make sure the Green wire is plugged in between the 1kΩ and 2kΩ resistors.
Camera shows black screen or says 'No device'	The ribbon cable is loose or inserted backwards.	Turn off the Pi, lift the camera clip, make sure the silver metal teeth face the HDMI ports, and push down firmly.
Pressing button does nothing	The button wire is on the wrong pin or disconnected.	Make sure the wire is on Physical Pin 11 and the other leg goes to Ground (Pin 9). Try pressing diagonally.
No sound comes out	The speaker is muted or set to wrong audio output.	Check speaker volume wheel. On the Pi, type: <code>espeak "test"</code> to confirm sound is working.
Pi suddenly restarts while walking	Weak power supply (not enough electrical juice).	Use the Official 27W Raspberry Pi power supply or a certified 5V/5A power bank. Avoid cheap 5W phone chargers.